

Performance of coherent audioband communication using consumer off the shelf material

Francesco Gramegna
School of computing
Korea Advanced Institute of Science and Technology
École Polytechnique Fédérale de Lausanne
francesco.gramegna@epfl.ch

Abstract—Most of the electronic devices we use every day have a built in microphone and speaker. In this research, we explore an implementation of TX-RX that allows communication using those devices. We focus our study on audible (0-20kHz) audioband communication, but also consider near-ultrasound (19kHz-24kHz) communication. We find that communication is possible at a data-rate of 1.5 kbits/s in audible range, and 50bit/s in near-ultrasound with our implementation.

I. INTRODUCTION

The implementation of a receiver and a transmitter using consumer off the shelf material (COTS) on audioband can be useful in many ways such as making everyday devices communicate between them without having to rely on Bluetooth (bypassing the OS). In this research, we explore an approach on how to implement a coherent communication receiver and transmitter, made for COTS.

We focus our implementation on audible communication, since it is the range where commercial microphones and speakers work best, however we will also consider near-ultrasound communication, since communicating at carrier frequency below 19kHz could disturb humans hearing the transmission.

All of the testing has been done in a real life environment (typically a furnished 4m*3m room with chairs and a table). The material used is a Iphone 12 mini for the speaker (TX), and a Lenovo Ideapad 5 for the microphone (RX). The final implementation of the TX-RX can be found at [1]

II. CHANNEL ANALYSIS

In this section we perform an analysis of the channel (Speaker - Air - Microphone).

A. Channel Gain

To get an estimate of the channel gain, we have transmitted a sine wave with continuous increasing frequency. The frequency is increased such that it goes from 0 Hz to 24000Hz in 60 seconds, i.e. every second the frequency increases by a total of 400Hz. We recorded the result, and calculated the approximated PSD over 1 second time frames. With the

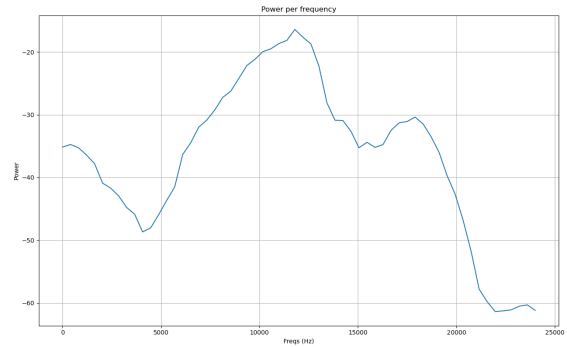


Fig. 1. Power received by frequency

approximated PSD, we have the approximate power of each 400Hz band-limited signals.

We plot in Figure 1 the power of each received frequency tone. For this plot we did not do any frequency synchronization, it is more of an informative plot of the power we get when we generate and receive a given frequency.

We see that the maximum gain is achieved in the range [10kHz, 13kHz]. We conclude that in environments where gaussian noise power is high, it is best to center our carrier around those values, to get the best possible energy per bit (E_b) and thus SNR .

We also notice that there is some gain around 20kHz (-40db) compared to the noise level of -60dB in 23kHz. This implies that near-ultrasound communication is possible with COTS.

B. Channel ISI

Since the performance of the communication system is evaluated indoors, we can expect the presence of inter symbol interference due to the channel impulse response.

In Figure 6 in the appendix we plot the recorded IR of two different rooms and see that the channel memory is about 30ms.

To get a grasp on the potential ISI, and to mitigate it, we have decided to perform MMSE equalization. For MMSE, we

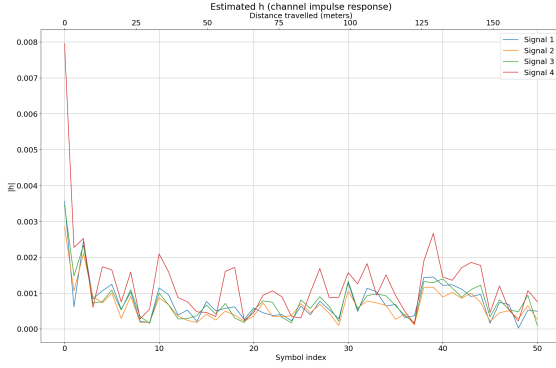


Fig. 2. Channel impulse response over symbols

need to have an estimate of the channel IR. To get the channel estimate we used the least-squares approach where we solve the following equation for h .

$$y[k] = \sum_i x[i]h[k-i] \quad (1)$$

x is the known preamble and y is the received message. TO solve this we transform the convolution in a matrix calculation by setting X as the Toeplitz of the preamble : $y = Xh$.

We then solve this using the pseudo-inverse of X : $h = X^{-1}y$. By doing this we get a channel impulse response per symbol. In Figure 2 we plotted such an impulse response. However we notice that our LS estimation might be a wrong method since the results we have imply that sound travels more than 150 meters in a room before fading away.

The reasons for the mathematical errors could be that the preamble is ill-formed, and creates an X matrix s.t. its pseudo-inverse is far from giving a good inverse estimate.

Given an erroneous channel estimate, it follows that MMSE channel equalization scrambles even more the received symbols. In Figure 8 in the appendix we see a typical reception of symbols without equalization, and in Figure 7 we have the reception with equalization. We see the direct performance reduction and thus decide not to put channel equalization in our communication system.

In the future we plan on implementing the communication system with an adaptive equalization system that does not require an explicit channel estimation.

C. Hardware processing

By using COTS hardware, we expose ourselves to the processing of incoming and outgoing data by the operating system, drivers and physical components.

Since some of these layers are proprietary (integrate microphone, or Apple OS), we have little knowledge of the processing they might do to our data.

For example, we infer that the built-in microphone in my computer might have an automatic gain control from Figure 3. We see that in the beginning of the preamble the signal is more distorted than in the end. However, the presence of these

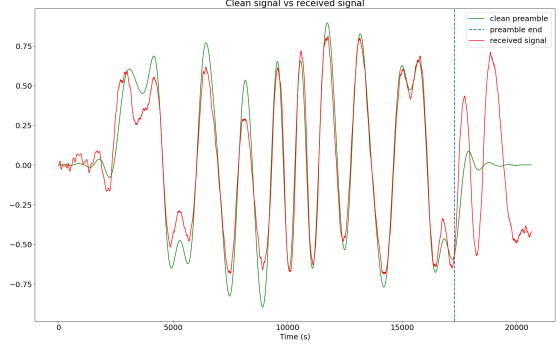


Fig. 3. ML Synchronization (real part of baseband)

hardware DSP modules is not really an issue: it seems that all of the components have linear phase frequency response. If not coherent communication would not have worked with our current implementation.

III. DESIGN CHOICES

A. Constellation

We decided to use QPSK, as 16-QAM gave us too high BER in our preliminary testing.

B. Synchronization

To use QPSK we needed to synchronize the phase, which we did using the *maximum likelihood* approach (ML).

We also needed to synchronize the timing. To do so we also used the ML approach.

The amplitude is normalized instead of synchronized, as it is not really important for QPSK.

In Figure 3 we see a timing and phase synchronization result.

C. Channel equalization

As discussed in Section II, we did not employ channel equalization in our final implementation.

D. Phase locked loop (PLL)

We noticed that there is frequency offset in higher f_c . In Figure 4 we plotted the received phase against the expected phase. We see that the phase slowly changes. In Figure 5, we instead see that the phase changes rapidly, indicating a high frequency offset.

We have implemented a PLL to mitigate this issue, and can see the result in green in the figures. However choosing the PLL parameters depended on the room we found ourselves in, and on the noise levels. This made it impractical to use a PLL in our final design, so we choose not to include it.

Instead we mitigate the issue by using packetized communication: since phase is synchronized from packet-to-packet, the transmitted and received phase offset errors are made negligible.

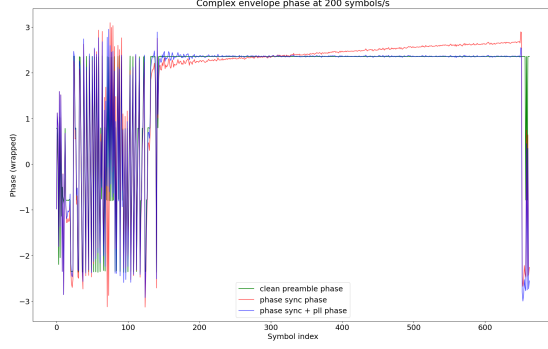


Fig. 4. Phase drift at $f_c = 1000\text{Hz}$

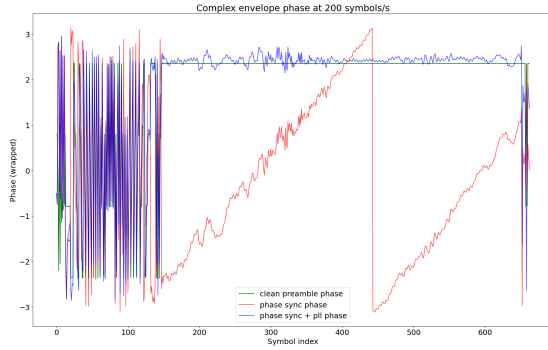


Fig. 5. Phase drift at $f_c = 18000\text{Hz}$

E. Channel coding

We did not implement any form of channel coding. The effective maximal bit/s under perfect channel coding can still be calculated:

$$R = R_e(1 - H_b(P_e)) \quad (2)$$

Where R_e is the bitrate with errors, P_e is the probability of error and H_b is the binary entropy.

This formula needs strong assumptions, s.t. our channel is symmetric. Nevertheless we keep this upper bound to compare the different parameters and results of our implementation in the section below.

IV. RESULTS

To evaluate the performance of our final implementation, we conducted 3 tests : One where the speaker (phone) was placed 10cm from the microphone (computer), another one where the transmitter was placed 1m from the receiver, and

a last test where we tested near-ultrasonic communication at 0.1m and 1m.

In the tables below, BER is the bit error rate, F_c is the carrier frequency, R_s is the symbol rate, P is the length of the preamble in symbols, W is the length of the payload in symbols, Bps is the bits per seconds calculated without the preamble, and the last column is the Max bps with no errors under optimal channel coding calculated with formula (2).

A. Ultra close range communication

TABLE I
PERFORMANCE AT 10CM OF DISTANCE

BER	F_c	R_s	P	W	Bps (b/s)	Max Bps no err
0.30058	1000	200	31	160	335.1	39.5
0.36442	10000	1000	227	800	1557.9	83.1

We notice in Table I that in extreme close range the performance is limited to only 83 bit/s without errors. This is probably due to heavy ISI.

B. Close range communication

TABLE II
PERFORMANCE AT 1M OF DISTANCE

BER	F_c	R_s	P	W	Bps (b/s)	Max Bps no err
0.32332	1000	200	31	160	335.1	30.9
0.08553	10000	2000	227	800	3115.8	1802.9
0.19028	10000	3000	227	1600	5254.5	1565.6
0.37440	12000	5000	551	2400	8132.9	374.3

We see from Table II that we have a significant performance increase when we increase the symbol rate. This is maybe due to more symbols arriving before major echo arrives from the distant wall in the room.

Our best performance is this 8% of BER, at around 3000bits/s. This performance gives us a peak at what our general performance could be if we implemented channel equalization, and did not have any ISI.

C. Near ultrasound communication

TABLE III
PERFORMANCE OF NEAR-ULTRASOUND COMMUNICATION (0.1M AND 1M)

BER	F_c	R_s	P	W	dist(m)	Bps (b/s)	Max Bps no err
0.2873	18kHz	500	127	160	0.1	557.5	75.1
0.2839	19kHz	500	63	160	0.1	717.5	100.0
0.3462	19kHz	500	127	240	1	654.0	45.3
0.4926	19kHz	500	127	380	1	749.5	0.11
0.1879	19kHz	100	7	32	1	164.1	49.7
0.3320	20kHz	100	7	32	1	164.1	13.6
0.1710	22kHz	100	7	32	1	164.1	55.8

We see from Table III the effect of the absence of a PLL in line 4 : we get significant degradation of performance when our payload size becomes larger. This is due to the phase changing too much before the end of the payload and resynchronization.

It is interesting to note that even though we are transmitting at frequencies above the human hearing range, a very faint static noise can be heard. The noise seems to be low frequency, so it might be due to a faulty speaker.

V. CONCLUSION

In this research, we explored how to implement an audio communication system that manages to send around 1800 b/s in audible audioband, and around 50 b/s in near ultra-sound audioband. We found that communication is possible without channel equalization under real world scenarios (however, as expected, it significantly decreases performance).

We found that we could communicate without implementing a PLL, greatly simplifying the receiver pipeline. To mitigate the issue of the PLL absence (and thus phase drift), we could also have used differential modulation. But for now making the communication windows smaller is a good enough solution.

An exciting discovery is the possibility of transmitting information between two every-day devices (an Iphone and a Laptop) in the near ultra-sound range, so without interfering with human communication.

REFERENCES

- [1] F. Gramegna, "Audioband RX TX implementation code." [Online]. Available: <https://github.com/francesco-gramegna/audioband-RXTX-demo> Accessed: Jan. 12, 2025.

APPENDIX

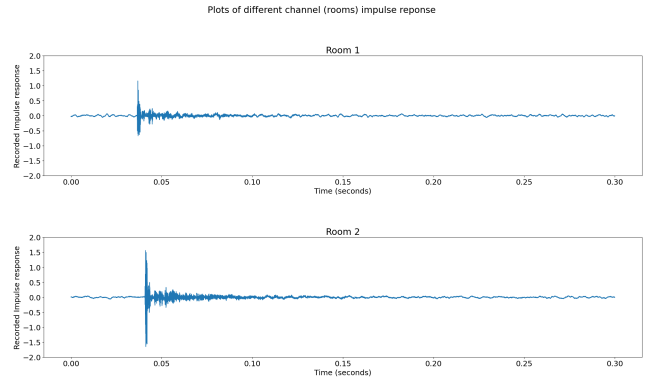


Fig. 6. IR of 2 different rooms

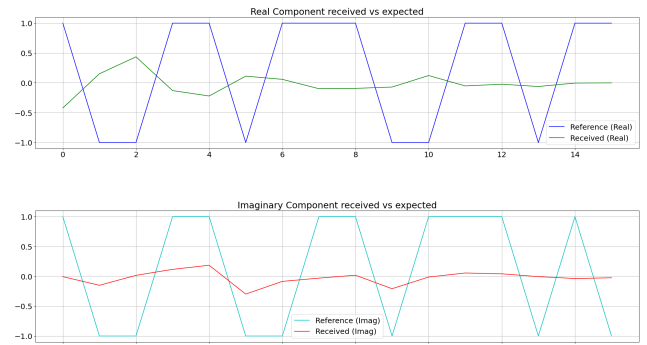


Fig. 7. Received symbols with equalization

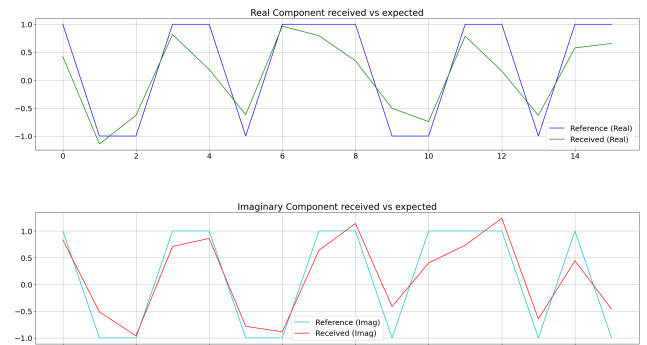


Fig. 8. Received symbols without equalization